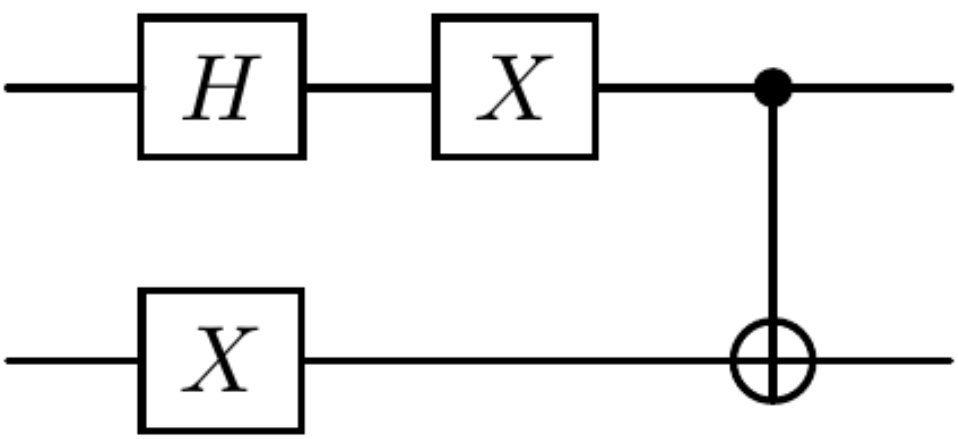
QuAIRKit

April 1, 2026

[1]:
```
import numpy as np
import torch
import quairkit as qkit
from quairkit import Circuit
from quairkit.database import *
from quairkit.operator import RZ, ChoiRepr, KrausRepr, StinespringRepr, Oracle
from quairkit.qinfo import *

import warnings
warnings.filterwarnings('ignore')
```

[4]:
```
num_qubits = 3  # number of qubits
cir = Circuit(num_qubits)
cir.h(0)  # Add Hadamard gate to qubit 0
cir.x([0, 1])  # Add X gate to qubit 0 and 1
cir.cx([0, 1])  # Add CNOT gate to qubit 0 and 1
print("the diagram of a quantum circuit:")
cir.plot()
print(f"the total number of qubit of the circuit is {cir.num_qubits}")
```

```
the diagram of a quantum circuit:
```



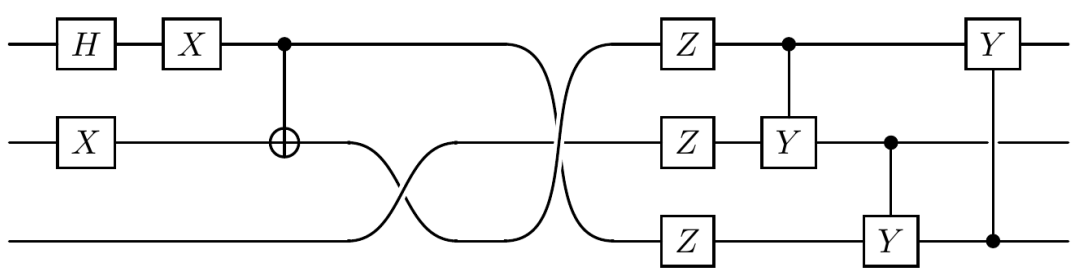
```
the total number of qubit of the circuit is 3
```

[5]:
```python
print(f"the unitary matrix of the circuit:\n{cir.matrix}")
```

```
the unitary matrix of the circuit:
tensor([[ 0.0000+0.j,  0.0000+0.j,  0.7071+0.j,  0.0000+0.j,  0.0000+0.j,
0.0000+0.j,
         -0.7071+0.j,  0.0000+0.j],
        [ 0.0000+0.j,  0.0000+0.j,  0.0000+0.j,  0.7071+0.j,  0.0000+0.j,
0.0000+0.j,
          0.0000+0.j, -0.7071+0.j],
        [ 0.7071+0.j,  0.0000+0.j,  0.0000+0.j,  0.0000+0.j, -0.7071+0.j,
0.0000+0.j,
          0.0000+0.j,  0.0000+0.j],
        [ 0.0000+0.j,  0.7071+0.j,  0.0000+0.j,  0.0000+0.j,  0.0000+0.j,
-0.7071+0.j,
          0.0000+0.j,  0.0000+0.j],
        [ 0.7071+0.j,  0.0000+0.j,  0.0000+0.j,  0.0000+0.j,  0.7071+0.j,
0.0000+0.j,
          0.0000+0.j,  0.0000+0.j],
        [ 0.0000+0.j,  0.7071+0.j,  0.0000+0.j,  0.0000+0.j,  0.0000+0.j,
0.7071+0.j,
          0.0000+0.j,  0.0000+0.j],
        [ 0.0000+0.j,  0.0000+0.j,  0.7071+0.j,  0.0000+0.j,  0.0000+0.j,
0.0000+0.j,
          0.7071+0.j,  0.0000+0.j],
        [ 0.0000+0.j,  0.0000+0.j,  0.0000+0.j,  0.7071+0.j,  0.0000+0.j,
0.0000+0.j,
          0.0000+0.j,  0.7071+0.j]])
```

[6]:
```python
cir.swap([[1, 2], [0, 2]])  # Add SWAP gate to qubit 1 and 2, and qubit 0 and 2
cir.z()  # Add Z gate to all qubits
cir.cy()  # Add CY gate in cycle
print("the diagram of the quantum circuit with more gates:")
cir.plot()
print(f"the total number of qubit of the circuit is {cir.num_qubits}")
```

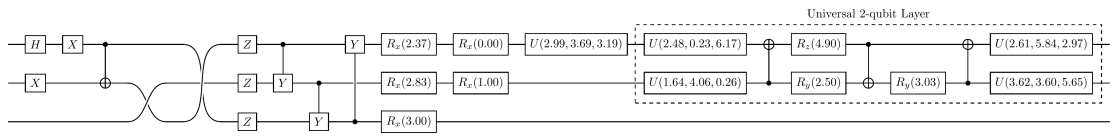
```
the diagram of the quantum circuit with more gates:
```



the total number of qubit of the circuit is 3

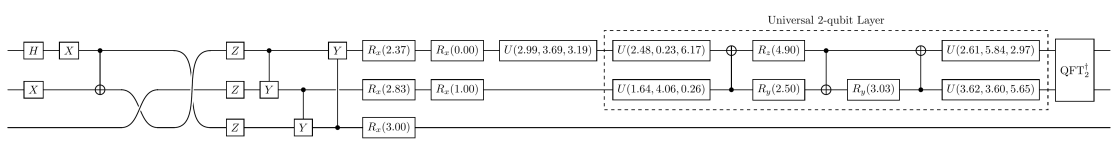
[7]:
```
cir.rx() # Add Rx gate to all qubits with random parameters
cir.rx([0, 1], param=np.arange(2)) # Add RX gate to the first two qubits with␣
↪specified parameters

cir.u3([0]) # Add universal single qubit gate
cir.universal_two_qubits([0, 1]) # Add universal two qubits gate
print("the diagram of a quantum circuit with parametrized gates:")
cir.plot()
```

the diagram of a quantum circuit with parametrized gates:



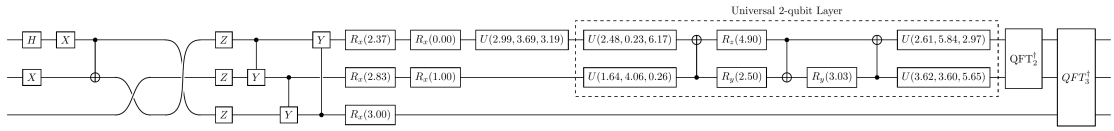
[8]:
```
qft_mat = qft_matrix(num_qubits=2)  # construct a 2-qubit quantum Fourier␣
↪transformation (QFT) operator
qft_inverse = dagger(qft_mat)  # get the inverse QFT

cir.oracle(qft_inverse, qubits_idx=[0, 1],␣
↪latex_name=r'\text{QFT}_{2}^{\dagger}')  # add custom gate
print("the diagram of a quantum circuit with a custom gate:")
cir.plot()
```

the diagram of a quantum circuit with a custom gate:



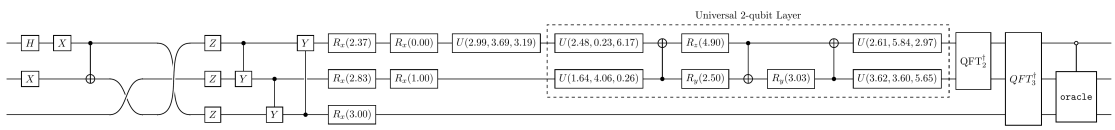
[9]:
```
qft_mat = qft_matrix(num_qubits=3)  # construct a 3-qubit quantum Fourier␣
↪transformation (QFT) operator
qft_inverse = dagger(qft_mat)  # get the inverse QFT
custom_unitary = Oracle(qft_inverse, system_idx=list(range(3)),␣
↪gate_info={"tex": r'QFT_{3}^{\dagger}'})
cir.append(custom_unitary)
print("the diagram of a quantum circuit with custom gates:")
cir.plot()
```

the diagram of a quantum circuit with custom gates:



[10]:
```
cir.oracle(random_unitary(num_qubits=2), qubits_idx=[0, 1, 2], control_idx=0,␣
↪gate_name="O")

print("the diagram of a quantum circuit with a controlled custom gate:")
cir.plot()
```

the diagram of a quantum circuit with a controlled custom gate:



[11]:
```
cir.depolarizing(prob=0.1, qubits_idx=[0])  # add a depolarizing channel (with␣
↪probability 0.1) to qubit 0
cir.bit_phase_flip(0.2)  # Add bit-phase flip channels (with probability 0.2)␣
↪to all qubits
rho = random_state(num_qubits=3)
replacement_choi_repr = replacement_choi(rho)  # replacement channel in Choi␣
↪representation
cir.choi_channel(replacement_choi_repr, system_idx=[0, 1, 2])  # add␣
↪replacement channel in Choi representation

reset_kraus_repr = reset_kraus([0.1, 0.2])  # reset channel in Kraus␣
↪representation
cir.kraus_channel(reset_kraus_repr, system_idx=2)  # add reset channel in Kraus␣
↪representation

random_stin_repr = random_channel(num_systems=1, target="stinespring")  #␣
↪random single-qubit channel in Stinespring representation
cir.stinespring_channel(random_stin_repr, system_idx=1)  # add random channel␣
↪in Stinespring representation
```

[12]:
```
choi_op = random_channel(num_systems=2, target="choi")
choi_repr = ChoiRepr(choi_op, system_idx=[1, 2])
cir.append(choi_repr)
```

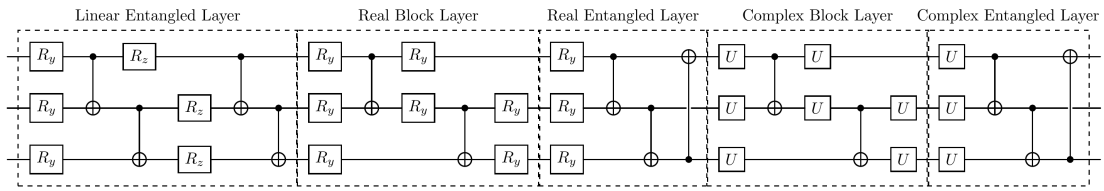
```
kraus_op = random_channel(num_systems=1, target="kraus")
kraus_repr = KrausRepr(kraus_op, system_idx=0)
cir.append(kraus_repr)

stine_op = random_channel(num_systems=3, target="stinespring")
stine_repr = StinespringRepr(stine_op, system_idx=[0, 1, 2])
cir.append(stine_repr)
```

[12]:
```
Circuit(
  (0): H()
  (1): X()
  (2): CNOT()
  (3): SWAP()
  (4): Z()
  (5): CY()
  (6): RX()
  (7): RX()
  (8): U3()
  (9): Universal2(
    (0): U3()
    (1): CNOT()
    (2): RZ()
    (3): RY()
    (4): CNOT()
    (5): RY()
    (6): CNOT()
    (7): U3()
  )
  (10): Oracle()
  (11): Oracle()
  (12): ControlOracle()
  (13): Depolarizing()
  (14): BitPhaseFlip()
  (15): ChoiRepr()
  (16): KrausRepr()
  (17): StinespringRepr()
  (18): ChoiRepr()
  (19): KrausRepr()
  (20): StinespringRepr()
)
```

[13]:
```
num_qubits = 3  # number of qubits
cir = Circuit(num_qubits)
cir.linear_entangled_layer(depth=1)  # add a linear entangled layer
cir.real_block_layer(depth=1)  # add real block layers in depth 2
cir.real_entangled_layer(depth=1)  # add a real entangled layer
```

```
cir.complex_block_layer(depth=1)  # add a complex block layer
cir.complex_entangled_layer(depth=1)  # add a complex entangled layer

print("the diagram of a quantum circuit with different kinds of layers:")
cir.plot(style='compact')
```

```
the diagram of a quantum circuit with different kinds of layers:
```



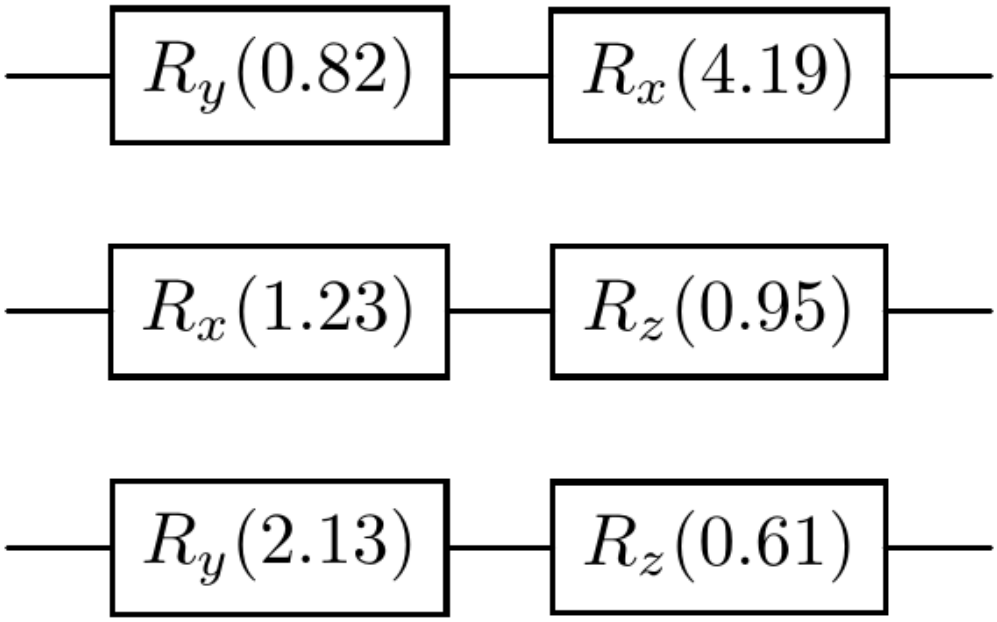
[14]:
```
print("The unitary matrix of the first layer is\n", cir[0].matrix)
```

```
The unitary matrix of the first layer is
 tensor([[ 0.0511-0.0331j, -0.0571+0.0369j,  0.5058-0.3272j, -0.5652+0.3656j,
         -0.0236+0.0153j,  0.0264-0.0171j, -0.2335+0.1510j,  0.2609-0.1688j],
        [ 0.0675-0.0083j,  0.0604-0.0075j,  0.6681-0.0826j,  0.5979-0.0739j,
         -0.0312+0.0039j, -0.0279+0.0034j, -0.3083+0.0381j, -0.2760+0.0341j],
        [-0.2807-0.5331j,  0.3136+0.5956j,  0.0284+0.0539j, -0.0317-0.0602j,
          0.1295+0.2460j, -0.1448-0.2749j, -0.0131-0.0249j,  0.0146+0.0278j],
        [-0.5420-0.3992j, -0.4851-0.3573j,  0.0548+0.0403j,  0.0490+0.0361j,
          0.2502+0.1843j,  0.2239+0.1649j, -0.0253-0.0186j, -0.0226-0.0167j],
        [ 0.0312+0.0039j,  0.0279+0.0034j,  0.3083+0.0381j,  0.2760+0.0341j,
          0.0675+0.0083j,  0.0604+0.0075j,  0.6681+0.0826j,  0.5979+0.0739j],
        [ 0.0236+0.0153j, -0.0264-0.0171j,  0.2335+0.1510j, -0.2609-0.1688j,
          0.0511+0.0331j, -0.0571-0.0369j,  0.5058+0.3272j, -0.5652-0.3656j],
        [-0.2502+0.1843j, -0.2239+0.1649j,  0.0253-0.0186j,  0.0226-0.0167j,
         -0.5420+0.3992j, -0.4851+0.3573j,  0.0548-0.0403j,  0.0490-0.0361j],
        [-0.1295+0.2460j,  0.1448-0.2749j,  0.0131-0.0249j, -0.0146+0.0278j,
         -0.2807+0.5331j,  0.3136-0.5956j,  0.0284-0.0539j, -0.0317+0.0602j]],
       grad_fn=<ViewBackward0>)
```

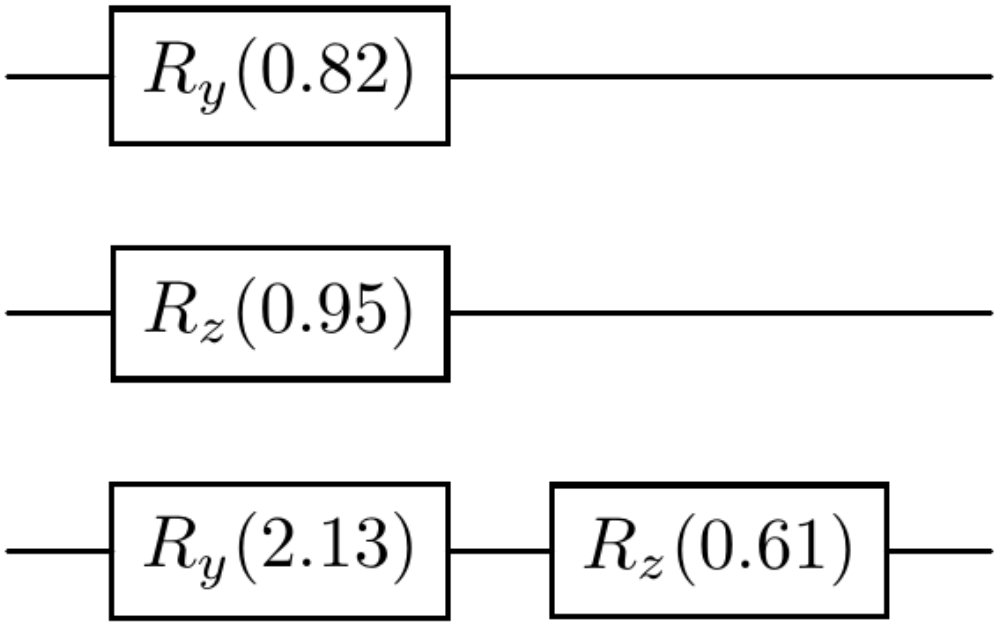
[15]:
```
cir = Circuit(num_qubits=3)
# Set parameters and customize quantum gates. Here we select Ry, Rx, Rz gates
param = np.random.rand(2)

# By default, randomly generates a set of parameters
rz_gate = RZ(param=param, qubits_idx=[1, 2])

# Add quantum gates
cir.ry([0, 2])
cir.rx([0, 1])
```

```
cir.insert(index=2, module=rz_gate)  # index where to insert

print("The quantum circuit after adding gates is: ")
cir.plot()

cir.pop(1)  # Remove Rx gate
print("The quantum circuit after removing gates is:")
cir.plot()
```

The quantum circuit after adding gates is:



The quantum circuit after removing gates is:



[16]:
```
output_state = cir()  # Run the circuit with initial state |0>
print("the output state for inputting zero state is:", output_state)

rho = random_state(num_qubits=3)
output_state = cir(rho)  # Run the circuit with initial state sigma
print("the output state for inputting state rho is:", output_state)
```

```
the output state for inputting zero state is:
-----------------------------------------------------
 Backend: default-pure
 System dimension: [2, 2, 2]
 System sequence: [1, 2, 0]
[0.32-0.31j 0.14-0.14j 0.79-0.13j 0.34-0.06j 0.  +0.j   0.  +0.j
 0.  +0.j   0.  +0.j  ]
-----------------------------------------------------

the output state for inputting state rho is:
-----------------------------------------------------
 Backend: default-mixed
 System dimension: [2, 2, 2]
 System sequence: [1, 2, 0]
[[ 0.17+0.j   -0.03+0.05j  0.04-0.01j  0.07-0.07j  0.02+0.06j -0.02-0.j
   0.05+0.12j  0.  -0.04j]
 [-0.03-0.05j  0.07+0.j    0.02-0.04j -0.05-0.05j -0.03-0.02j  0.01-0.02j
   0.04-0.07j -0.06-0.02j]
```

```
 [ 0.04+0.01j  0.02+0.04j  0.09+0.j    0.02-0.04j -0.01-0.05j  0.03-0.03j
   0.11+0.02j -0.02-0.01j]
 [ 0.07+0.07j -0.05+0.05j  0.02+0.04j  0.18-0.j    0.03+0.01j -0.02+0.01j
  -0.02+0.1j   0.1 -0.03j]
 [ 0.02-0.06j -0.03+0.02j -0.01+0.05j  0.03-0.01j  0.09+0.j   -0.  +0.04j
   0.04+0.09j  0.  +0.01j]
 [-0.02+0.j    0.01+0.02j  0.03+0.03j -0.02-0.01j -0.  -0.04j  0.04-0.j
   0.04+0.01j -0.01-0.02j]
 [ 0.05-0.12j  0.04+0.07j  0.11-0.02j -0.02-0.1j   0.04-0.09j  0.04-0.01j
   0.24+0.j   -0.04-0.j  ]
 [ 0.  +0.04j -0.06+0.02j -0.02+0.01j  0.1 +0.03j  0.  -0.01j -0.01+0.02j
  -0.04+0.j    0.12+0.j  ]]
-----------------------------------------------------
```

[17]:
```python
print("the circuit depth is", cir.depth, "\n")

print("the gate history of the circuit is\n", cir.operator_history)
```

```
the circuit depth is 2

the gate history of the circuit is
 [{'uid': '9affe910-10fe-4497-b93f-7692cc1626b3', 'name': 'ry', 'type': 'gate',
'tex': 'R_{y}', 'api': 'ry', 'param_sharing': False, 'plot_width': 0.9,
'system_idx': [[0], [2]], 'param': Parameter containing:
tensor([[[0.8233]],

        [[2.1282]]], requires_grad=True), 'matrix': tensor([[[ 0.9165+0.j,
-0.4001+0.j],
         [ 0.4001+0.j,  0.9165+0.j]],

        [[ 0.4853+0.j, -0.8743+0.j],
         [ 0.8743+0.j,  0.4853+0.j]]], grad_fn=<SqueezeBackward0>)}, {'uid':
'c6864dc4-2228-4f97-9020-e2830f02a2aa', 'name': 'rz', 'type': 'gate', 'tex':
'R_{z}', 'api': 'rz', 'param_sharing': False, 'plot_width': 0.9, 'system_idx':
[[1], [2]], 'param': tensor([[[0.9483]],

        [[0.6101]]]), 'matrix': tensor([[[0.8897-0.4566j, 0.0000+0.0000j],
         [0.0000+0.0000j, 0.8897+0.4566j]],

        [[0.9538-0.3003j, 0.0000+0.0000j],
         [0.0000+0.0000j, 0.9538+0.3003j]]])}]
```

[18]:
```python
print("the trainable parameters of entire circuit are", cir.param)

cir.update_param(torch.ones_like(cir.param))  # update the parameters of the␣
 ↪circuit
print("the updated trainable parameters of entire circuit are", cir.param)
```

```
the trainable parameters of entire circuit are tensor([0.8233, 2.1282])
the updated trainable parameters of entire circuit are tensor([1., 1.])
```

[19]:
```
qkit.print_info()
```

```
---------VERSION---------
quairkit: 0.4.4
torch: 2.11.0+cu130
torch cuda: 13.0
numpy: 2.4.3
scipy: 1.17.1
matplotlib: 3.10.8
---------SYSTEM---------
Python version: 3.13.9
OS: Linux
OS version: #107-Ubuntu SMP PREEMPT_DYNAMIC Fri Mar 13 19:51:50 UTC 2026
---------DEVICE---------
CPU:  Intel(R) Core(TM) i7-4770 CPU @ 3.40GHz
GPU: (0) NVIDIA GeForce GTX 1650
```

[ ]:
```

```